

Yan Wang

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EDUCATION

Wuhan University

Wuhan, China

- B.S. in Biological Science; **GPA: 3.91/4.00; Major GPA: 3.97/4.00; Ranked #1** in the major Sep 2023- Jun 2027
- **Hongyi Honors Class:** University-Wide Selective Cohort For Advanced Research Training
- **Yale School of Medicine: Senior Thesis** Jul 2026 – May 2027

SCHOLARLY OUTPUTS

Manuscripts Under Review

Wang, Y., Miller, L., Estrada, M., & Li, D. Mitochondrial dysfunction and α -synuclein pathology in Parkinson's disease: A bidirectional pathogenic loop and therapeutic opportunities. *Frontiers in Molecular Neuroscience*. Review manuscript under review.

Conference Abstracts

Wang, Y., Zhang, J., Huang, X., Ruan, Y., & Yu, Y. Distributed INS-1 signaling through the candidate noncanonical receptor NTR-1 tunes the ASER – AIY axis to drive high-salt ethanol avoidance in *Caenorhabditis elegans*. Abstract accepted for presentation at *Neuroscience 2026, Society for Neuroscience*.

Grants & Funding

Young Talent Sponsorship Program, Unitree Robotics. Project: “Multimodal monitoring of reversible hypometabolic states.” Application under review (anticipated decision Sept 2026).

RESEARCH EXPERIENCE

Adams Center for Parkinson's Disease Research, Yale School of Medicine

New Haven, CT

Senior Thesis, Visiting Undergraduate Researcher | Mentor: Prof. Clemens Scherzer and Dunhui Li

Jul 2026-Present

Project: Functional Dissection of the PM20D1 Locus in Parkinson's Disease

- Investigate how *PM20D1* perturbation affects mitochondrial function and cellular redox homeostasis in human iPSC models using antisense oligonucleotides (ASOs) and gene-editing approaches.
- Characterize the subcellular localization of the lncRNA *PM20D1-AS1* and investigate its potential regulation of *PM20D1* expression and signal pathway, while exploring candidate molecular interaction partners.
- Independently develop RNA-seq workflows to profile *PM20D1-AS1* transcript isoforms, identifying PD-associated isoform shifts in astrocytes and oligodendrocytes and investigating how disease-associated SNPs alter splicing to uncover potential genetic mechanisms underlying Parkinson's disease.

Medical Research Institute, Wuhan University

Wuhan, China

Undergraduate Researcher; Project I Lead | Joint lab of Prof. Yanxun Yu and Prof. Younghan Jin

Jan 2025-Jul 2026

Project I: INS-1-Dependent Neuropeptidergic Tuning of Sensory Valence in *C. elegans*

- Independently elucidated the mechanism underlying the *ins-1*-dependent reversal of high-salt ethanol avoidance into attraction.
- Mapped distributed INS-1-releasing nodes in *C. elegans* by combining cell-specific *ins-1* deletion, rescue, and neuronal ablation, implicating ASEL, AIY, AIZ, and AIB in ethanol avoidance.
- Demonstrated ASER-AIY circuit dysfunction using calcium imaging, showing a right-shifted ASER ethanol response range and weakened AIY negative calcium transients in *ins-1* mutants.
- Revealed ASEL-derived INS-1 as a major regulator of ASER sensory tuning. Integrated ethanol dose-dependent behavioral assays with physical modeling of diffusion, supporting a quantitative model in which INS-1 tunes ethanol sensitivity rather than causing simple sensory loss.
- Proposed NTR-1 as a candidate noncanonical INS-1 receptor by integrating pathway mutant analysis, *ntr-1* phenotyping, molecular docking and MD simulations.
- Spearheaded a circuit-level model in which distributed INS-1 release, potentially mediated by NTR-1, calibrates ASER sensory gain and downstream AIY output to code ethanol valence under high-salts.

Project II: Performance Optimization of the POST-IT Proximity Tagging System

- Optimized POST-IT, a Pupylation-based non-diffusive proximity-labeling platform for live-cell-compatible small-molecule target identification.

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- Engineered PafA ligase to reduce the major kinetic and substrate constraints of the original POST-IT system, including long labeling time and high SBP-Pup (E) requirement.
- Built a yeast-surface-displayed Pep1-MATH weak-interaction selection system to couple PafA catalytic activity with FACS-based enrichment of improved variants.
- Executed iterative directed evolution of a $\sim 10^6$ -scale PafA mutant library constructed via error-prone PCR, integrating FACS enrichment with NGS-guided mutation tracking under progressively stringent labeling conditions to isolate high-performing variants.
- Improved labeling from 18h/40 μ M to 2h/10 μ M, increasing threshold-positive cells from 0.5% to 27.2% and achieving >40-fold apparent performance enhancement.

College of Life Science, Wuhan University

Wuhan, China

Wet Lab Leader; Project Prototype Designer, 2025 iGEM Gold Medal | Advisor: Assoc. Prof. Lei Bai

Nov 2024-Oct 2025

Project: LOGIC -Lasting Optimized Genetic Input Calculator

- Originated and designed LOGIC, utilizing quorum-sensing-mediated intercellular communication as programmable spatial “wiring” for colony-level bio-computation.
- Designed a diffusion-based biocomputing architecture in which orthogonal AHL signaling channels, engineered sender-receiver colonies, and spatial concentration gradients are integrated to implement modular genetic logic beyond single-cell circuit constraints.
- Built and characterized high-pass and band-pass response modules to encode Boolean operations, construct AND, OR, XOR, multi-input gates, and cascaded logic operations on solid media.
- Developed a serial biological adder framework by coordinating input, output, and carry signals, advancing the system from isolated logic gates toward executable biological arithmetic.
- Integrated optogenetic refresh modules, including split-VVD AiiA for light-controlled AHL degradation and LOVdeg-tagged output proteins for signal reset.

TECHNICALSKILLS

Model Organisms & Cell Culture: *E. coli*, *S. cerevisiae*, *C. elegans* manipulation; human iPSC culture and maintenance

Molecular & Biochemical Methods: Molecular cloning; RNA techniques (e.g., qPCR, RNAscope, RNA pull-down); protein expression and purification; Western blot; co-immunoprecipitation; FACS screening; general biochemical assays

Neuroscience & Imaging Techniques: *C. elegans* — neuronal calcium imaging, optogenetic manipulation, microinjection, genetic crossing, and behavioral phenotyping; confocal microscopy

Computational & Structural Biology: snRNA-seq data analysis; customized Python/R analytical pipelines; molecular docking; molecular dynamics (MD) simulations

HONORS & AWARDS

Hongyi Honors College's Annual Top Ten Student Innovators Award (¥10,000) Sep 2026

Westlake University Rising Star Award (¥30,000), top 9 out of 300 candidates Aug 2026

Liu Daoyu Creative Learning Scholarship (¥8,000), awarded to top 24 university-wide students Jan 2026

Gold Medal, International Genetically Engineered Machine Competition (**iGEM**) Nov 2025

Multiple Gold Prizes, Regular Track, Synthetic Cell Track, Protein Design Track, Bacterial Game Track | **SynBio Challenges** (International Synthetic Biology Competition) 2025–2026

Hongyi Honors College Annual Top Ten Science Advances Award (¥10,000) Oct 2025

Outstanding Student Scholarship (top 5% cohort), Outstanding Student & Outstanding Student Leader 2024–2025

First-Class Outstanding Freshman Scholarship 2023

ACADEMIC & SCHOLARLY TRAINING

Participant, Neuroscience & Cognitive Science Summer Training Program, **Peking University** Jul 2026

Participant, “Frontiers of Life Sciences” International Summer School, **Westlake University** Jul–Aug 2025

Delegate, National “Top-Tier Talent 2.0” Biology Forum, **Tsinghua University** May 2025